

IntelliCane: One-Page Technical Proposal

Smart Assistive Cane with Haptic Obstacle Avoidance, Fall Detection, and a Mobile App for Blind Users and their Caretakers

Team Neural-Nexus | Track A: Embedded IoT System | IoTrix 2.0 Semi-Final (5 Oct 2026)

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Repository Link: <https://github.com/Neural-Nexus-IoTrix-2-0/intellicane>

1. Problem Statement

Visually impaired individuals face persistent safety hazards from undetected low-hanging or ground-level obstacles and emergency falls. Existing Electronic Travel Aids (ETAs) often exceed \$500 USD (LKR 150,000+), rely on continuous cloud connectivity for basic hazard detection, and emit overwhelming audio spam that obstructs natural hearing. There is an urgent need for an affordable, offline-safe smart cane providing intuitive tactile guidance and caretaker emergency bridging.

2. Proposed Solution

IntelliCane is a low-cost (LKR 2,800 bench electronics), multimodal mobility aid combining an offline embedded safety reflex on the cane with an accessible companion Android application (`com.example.intellicane`):

- Offline Safety Reflex (ESP32-C3):** Sub-50ms deterministic loop. An HC-SR04 ultrasonic transceiver detects forward hazards up to 4m, driving a 3-pin vibration motor via 200Hz LEDC PWM that scales smoothly between 60cm and 10cm. An MPU6050 6-axis IMU detects orientation/impact ($> 65^\circ$ tilt or $> 2.5g$). Upon fall detection, the cane automatically enters a silent emergency state (disabling the local buzzer and motor to prevent user distress) and immediately broadcasts high-priority alert packets over BLE.
- Android Mobile App (application/):** Connects over BLE 5.0 (Nordic UART Service) and features two tailored modes: *Blind User Mode* (accessible dashboard, one-tap location sharing, collapsible floating terminal for live diagnostics) and *Caretaker Mode* (real-time presence status, Firebase Firestore fall synchronization, and an interactive ESRI/OSMDroid map with a dedicated user-location re-centering FAB).

3. System Architecture & Data Flow

Sensing (Cane) HC-SR04 (Trig: 0, Echo: 1 direct) MPU6050 6-Axis IMU (I2C: 4/5)	→	Core Embedded Reflex (ESP32-C3) 160 MHz RISC-V, Non-blocking scheduler Monotonic ramping math & fall debounce	→	Tactile & Audio Alerts 200 Hz LEDC PWM Haptic Motor (GPIO 6) Proximity Audio Warning (Silent on Fall)
Wireless BLE 5.0 Nordic UART Service (RX: 6E400002 / TX: 6E400003)				
Mobile App (Kotlin) User Dashboard & Caretaker Modes Floating Terminal & BLE Client	→	Location & Cloud Services Smartphone A-GPS Geolocation Firebase Firestore Sync & Presence	→	Caretaker View Real-time OSMDroid/ESRI Map (FAB Centering) Remote Fall Alert Push & Presence

4. Current Progress & Verification

- Physical Hardware Bench Build:** Assembled and verified with ESP32-C3 SuperMini, direct-wired HC-SR04 Echo (GPIO 1), GY-521 MPU6050, 3-pin vibration motor, and buzzer.
- Proportional Haptics:** 200 Hz LEDC PWM verified ramping smoothly from 60 cm down to 10 cm, remaining silent > 60 cm to prevent sensory fatigue.
- Silent Fall Detection & BLE Dispatch:** Tilt $> 65^\circ$ or impact $> 2.5g$ silences local actuators and dispatches `FALL.STATE:1` over BLE; upright recovery ($< 30^\circ$) transmits `FALL.STATE:0` and restores navigation.
- Resilient Multi-IC Driver:** Custom direct-register fallback driver supports clone MPU6050/6500 ICs, dual I2C addresses (0x68/0x69), and bus auto-reconnect.
- Wireless BLE Telemetry:** 10 Hz live packet broadcast over Nordic UART Service displaying connected smartphone peer MAC address.
- Android Application Implemented & Refined:** Native Kotlin app (`com.example.intellicane`) with User Dashboard, floating collapsible terminal window, Firebase Firestore fall sync (`updateFallStateInFirestore`), and live map with user-centering FAB (`fabUserLocation`).

5. Technology Stack & Budget Breakdown (Phase 1 Bench: LKR 2,800)

- Hardware:** ESP32-C3 SuperMini (LKR 1,400), HC-SR04 Ultrasonic (LKR 450), GY-521 MPU6050 IMU (LKR 600), 3-Pin Vibration Motor (LKR 250), Piezo Buzzer & Jumpers (LKR 100). **Total: LKR 2,800** (Tracked within the LKR 20,000–35,000 competition budget ceiling).
- Software & Protocols:** C++17, Arduino-ESP32, Kotlin Android, Firebase, OSMDroid, I2C (100 kHz), BLE 5.0 GATT NUS (10 Hz).

6. Testing & Validation Evidence

- Host Unit Testing (tests/proximity_feedback_test.cpp):** 100% pass across boundary inputs (negative, NaN, ∞), monotonic PWM progression, motor polarity inversion, beep transitions, and 32-bit `millis()` rollover resilience.
- Bench Calibration:** HC-SR04 verified from 2 cm to 250 cm with ± 1.5 cm accuracy. Tilt alarm calibrated with digital protractor ($> 65^\circ$ trigger, $< 30^\circ$ clear).
- BLE Throughput & Thermals:** Telemetry stream validated at 10 Hz with < 25 ms latency over 10 m range. Electronics maintained $< 32^\circ\text{C}$ under 100% motor duty.

7. Limitations & Risk Management

- Ultrasonic Specular Scattering:** Sound waves can scatter on angled soft fabrics beyond 2 m (*Mitigation:* Adding Time-of-Flight laser sensor for the final).
- Jumper Durability:** Prototype uses breadboard jumpers (*Mitigation:* Transitioning to a custom 2-layer PCB and 3D-printed handle for final).
- Phone Dependency for Cloud:** If phone battery dies, cloud tracking pauses (*Mitigation:* Core obstacle avoidance and fall buzzer remain 100% functional offline).

8. Roadmap Prior to the Final (17 October 2026)

- Custom PCB & 3D Handle:** Fabricate a custom PCB and 3D-print an ergonomic handle housing with TPU vibration dampening.
- End-to-End Caretaker Cloud Flow:** Finalize Firebase Cloud Functions to dispatch automated SMS and push notifications on fall confirmation.
- Battery Optimization:** Integrate an 18650 Li-ion cell with TP4056 charge controller and implement ESP32-C3 Light Sleep for > 24 hour runtime.